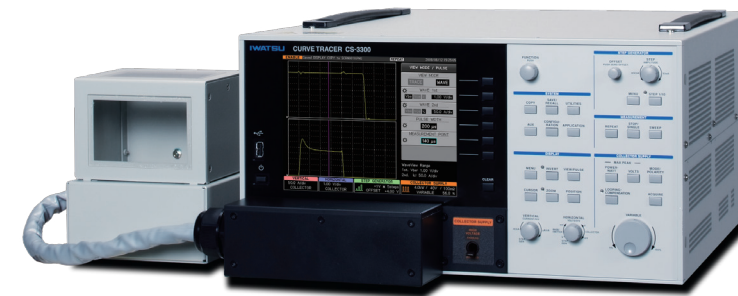


Semiconductor Curve Tracer Series

suitable for various semiconductor I-V curve measurement
such as IGBTs, MOS-FETs, Transistors, Diodes, etc

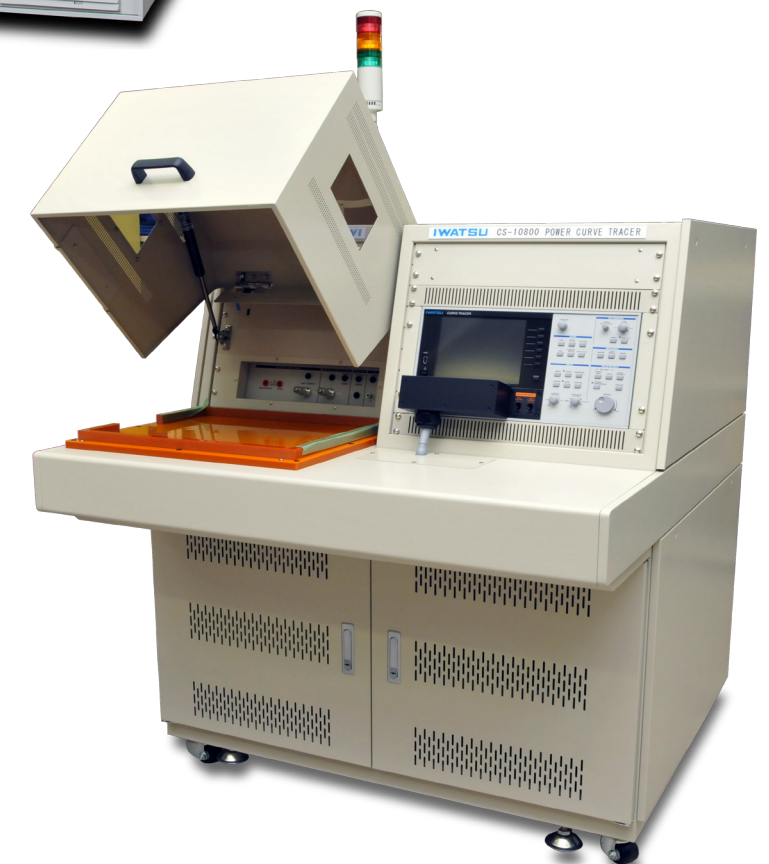


Curve Tracer model :
CS-5100
CS-3100



Curve Tracer model :
CS-5300
CS-5200
CS-3300
CS-3200

Curve Tracer model :
CS-10800
CS-10400



Semiconductor Curve Tracer Series Specifications

Collector supply		CS-3100	CS-3200	CS-3300	CS-5100	CS-5200	CS-5300	CS-10400	CS-10800										
Mode	High voltage1	AC, ± Rectified SIN, ± DC, ± LEAKAGE																	
	High voltage2	-						+DC											
	High current	-	±Pulse		-	±Pulse		±Pulse											
Maximum peak power	High voltage1	120mW, 1.2W, 12W, 120W, 390W ※390 W can be selected excluding the maximum peak voltage setting at 3,000V range.			120mW, 1.2W, 12W, 120W, 390W ※ 120mW and 390 W can be selected excluding the maximum peak voltage setting at 5,000V range.			120mW, 1.2W, 12W, 120W, 390W ※390 W can be selected excluding the maximum peak voltage setting at 3,000V range.											
	High voltage2	-						40W, 400W, 4kW											
	High current	-	400W, 4kW	400W, 4kW, 10kW		-	400W, 4kW	400W, 4kW, 10kW	600W, 6kW, 60kW	600W, 6kW, 60kW, 64kW									
High voltage mode 1		Maximum peak current(maximum peak pulse current)		Maximum peak voltage		Maximum peak current(maximum peak pulse current)		Maximum peak voltage		Maximum peak current(maximum peak pulse current)		Maximum peak voltage							
		75mA(150mA)		3,000V(2,500V / AC)		25mA(25mA)		5,000V(2,500V / AC)		75mA(150mA)		3,000V(2,500V / AC)							
		750mA(1.5A)		300V		750mA(1.5A)		300V		750mA(1.5A)		300V							
		7.5A(15A)		30V		7.5A(15A)		30V		7.5A(15A)		30V							
Loop compensation		The floating capacitance between the collector terminal and the ground of the fixture is compensated by the hardware in the high voltage mode. Digital compensation is also available.																	
High voltage mode2		-								Maximum peak current		Maximum peak voltage							
										400mA		10,000V							
High current mode (pulse only)		-	Maximum peak current		Maximum peak voltage		-	Maximum peak current		Maximum peak voltage		Maximum peak current		Maximum peak voltage		Maximum peak current		Maximum peak voltage	
			400A		40V			400A		40V		4,000A		60V		8,000A		60V	
			40A		40V			400A		40V		400A		60V		4,000A		60V	
								40A		40V		40A		60V		400A		60V	
												40A		60V		40A		60V	
Pulse width / measurement point		-	The pulse width is variable in the range from 50μs to 400 μs. Measurement points can be specified (with a resolution of 10 μs/step).			-	The pulse width is variable in the range from 50μs to 400 μs. Measurement points can be specified (with a resolution of 10 μs/step).			The pulse width is variable in the range from 50μs to 400 μs. (50μs to 1,200 μs / over 4kV) Measurement points can be specified (with a resolution of 10 μs/step).									
Maximum data points		20 to 1,000 points can be specified per trace.																	

Step generator

Current mode	Amplitude range: 50 nA to 200 mA, 21 ranges, 1-2-5 step Maximum current: ±2A / Offset: ± 10 times STEP AMPLITUDE setting			
Voltage mode	Amplitude range: 50 mV to 2 V, 6 ranges, at 1-2-5 step Maximum voltage:±40V / Offset: ± 10 times STEP AMPLITUDE setting			
Step rate	Staircase wave: 2times of 50 Hz or 60 Hz (50 Hz or 60 Hz in the AC mode)	Staircase wave: 2times of 50 Hz or 60 Hz (50 Hz or 60 Hz in the AC mode) Pulse : The pulse changes in the range from 80ms to 1,000 ms. (The lowest frequency limited by the maximum peak power consumption setting.)	Staircase wave: 2times of 50 Hz or 60 Hz (50 Hz or 60 Hz in the AC mode)	Staircase wave: 2times of 50 Hz or 60 Hz (50 Hz or 60 Hz in the AC mode) Pulse : The pulse changes in the range from 80ms to 1,000 ms. (The lowest frequency limited by the maximum peak power consumption setting.)
Pulse step	The pulse width is variable in the range from 50 μs to 400 μs at 10 μs resolution.			
Number of steps	0 to 20steps			

AUX output

Range	OFF, -40 to +40 V, variable with at 100mV resolution
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Vertical axis

Collecto current	High voltage	1 μA/div to 2 A/div, 20 steps, 1-2-5 switching						
	High current	-	100 mA/div to 50 A/div, 9 ranges, at 1-2-5 step	100 mA/div to 100 A/div, 10 ranges, at 1-2-5 step	-	100 mA/div to 50 A/div, 9 ranges, at 1-2-5 step	100 mA/div to 100 A/div, 10 ranges, at 1-2-5 step	100 mA/div to 1,000 A/div, 13 ranges, at 1-2-5 step
Emitter current(LEAKAGE)		1 nA/div - 2 mA/div, 20 ranges, at 1-2-5 steps						

Horizontal axis

Collector voltage	High voltage	50 mV/div to 500 V/div, 13 ranges, at 1-2-5 step						50 mV/div to 1,000 V/div, 14 ranges, at 1-2-5 step
	High current	-	50 mV/div to 5 V/div, 7 ranges, at 1-2-5 step	-	50 mV/div to 5 V/div, 7 ranges, at 1-2-5 step	-	50 mV/div to 10 V/div, 8 ranges, at 1-2-5 step	
Base / emitter voltage		50 mV/div to 5 V/div, 7 ranges, at 1-2-5 step						

Miscellaneous

Display	8.4-inch color TFT-LCD (SVGA 800 x 600 pixels)
Data save / recall	Internal: Memory (setup: 256, REF waveform: 4) External: Removable storage connected to the USB port (setup, waveform save / recall, screen hard copy)
USB	1 port (USB1.1)
Remote control	Remote control by LAN: 1 port (100BASE-TX)

Mechanism section

Dimensions (mm)	424 (W) x 555.2 (L) x 221 (H) (without the external projection portion)	424 (W) x 555.2 (L) x 354.5 (H) (without the external projection portion)	424 (W) x 555.2 (L) x 221 (H) (without the external projection portion)	424 (W) x 555.2 (L) x 354.5 (H) (without the external projection portion)	1,110 (W) x 1,150 (L) x 1,216 (H) (without the external projection portion)
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